

Fast VQF: Speed-Optimized 9DOF AHRS via Euler Integration and Combined Sensor Correction

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Abstract—This paper presents Fast VQF, a computationally optimized Attitude and Heading Reference System (AHRS) filter for 9-degree-of-freedom (9DOF) inertial sensor fusion. The proposed method builds on the VQF algorithm and introduces two targeted optimizations: (1) Euler-method quaternion integration replacing the trigonometric axis-angle propagation, and (2) a single combined accelerometer–magnetometer correction step replacing VQF’s two sequential corrections. The Euler integration eliminates the $\cos/\sin$ calls in gyroscope propagation, contributing a 5 % reduction in total execution time. The combined correction eliminates the atan2 computation and an intermediate quaternion state, contributing a further 44 % reduction. Together, the two optimizations achieve approximately $1.9\times$ speedup over Basic VQF ($3.8\times$ over full VQF) on a desktop CPU, with larger gains expected on microcontrollers where trigonometric functions are computed in software. Results on the BROAD [1] and Sassari [2] public benchmark datasets show that Fast VQF achieves the lowest orientation RMSE of all tested methods on undisturbed motion (2.62°), outperforming Basic VQF (2.71°), full VQF (2.66°), Justa AHRS [3] (2.84°), and Madgwick [4] (2.87°). Under disturbed conditions, full VQF’s bias estimation and disturbance rejection provide a clear advantage. The results confirm that modern fast methods are preferable to Madgwick for undisturbed applications, while full VQF remains the best choice under harsh magnetic conditions. Handling of optical reference discontinuities in the benchmark is also addressed.

I. INTRODUCTION

Attitude and Heading Reference Systems (AHRS) fusing gyroscope, accelerometer, and magnetometer data are central to inertial navigation, robotics, and motion analysis. Applications ranging from wearable sensors to unmanned aerial vehicles require both accuracy and computational efficiency, particularly on embedded processors or when computational latency is critical (pose prediction).

Classical complementary filters—Mahony [5] and Madgwick [4]—offer low computational cost but limited accuracy under dynamic conditions. The VQF (Versatile Quaternion-based Filter) algorithm [6] significantly raised the accuracy bar, achieving a mean RMSE of 2.9° across a diverse collection of datasets and hardware platforms with a single fixed parametrisation, compared to 5.3° – 16.7° for eight competing methods. Since VQF was already thoroughly validated against a wide set of competing algorithms in [6], the present work uses it as the single comparison baseline rather than repeating that survey. VQF filters the accelerometer in the near-earth frame and applies two sequential corrections (tilt via accelerometer, heading via magnetometer), exposing three quaternion states at each step. The gyroscope propagation uses the axis-angle formula, requiring $\cos$ and $\sin$ evaluations. The heading correction requires atan2 . Both are relatively costly

on microcontrollers without dedicated floating-point hardware, and all three correction stages must run at every time step for any application requiring 9DOF output.

This work introduces Fast VQF, which applies two specific structural optimizations to basicVQF: Euler-method quaternion integration (Optimization 1) and a combined single-step accelerometer–magnetometer correction (Optimization 2). Together these yield $\approx 1.9\times$ speedup over basicVQF; on undisturbed data Fast VQF achieves the lowest RMSE of all evaluated methods, while under disturbed conditions the additional complexity of full VQF provides clear accuracy benefits. We also identify and address data quality issues in the BROAD benchmark: the optical tracking reference contains sporadic discontinuities.

Contributions:

- Euler-method quaternion integration eliminating $\cos/\sin/\sqrt{\cdot}$ from gyroscope propagation: 15 % faster on the integration step (5 % overall), with no measurable accuracy impact across all tested datasets. The per-step gain is larger on microcontrollers where transcendental functions are software-computed.
- Combined single-step accelerometer–magnetometer correction eliminating atan2 and an intermediate quaternion state, contributing 44 % of total execution time reduction.
- Evaluation on BROAD [1] and Sassari [2] benchmark datasets with explicit handling of optical reference discontinuities and magnetic disturbance segments.

II. RELATED WORK

VQF [6] (Laidig and Seel, 2023) filters accelerometer measurements in the near-earth frame to suppress dynamic acceleration artefacts. It maintains three sequential correction stages: gyroscope integration via axis-angle, accelerometer-based tilt correction, and magnetometer-based heading correction. Gyroscope bias estimation and magnetic disturbance rejection are included in the full variant; a stripped basicVQF variant omits these. Its open-source implementation and broad evaluation make it the primary baseline of the present work.

Justa AHRS [3] introduced an AHRS predictor-corrector framework with independent sensor correction steps. The magnetometer correction formulation from that work is directly inherited here. Two variants were proposed: $\text{SO}(3)$ -based for accuracy and linearized for speed, both outperforming the state of the art at the time.

Mahony [5] and **Madgwick** [4] filters remain widely cited for lightweight orientation estimation. They apply proportional-integral feedback and gradient-descent correction

respectively. Both are today outperformed by modern methods such as VQF under varied dynamic conditions, and their accuracy is insufficient for demanding applications.

III. PROPOSED ALGORITHM

A. Notation

Unit quaternions $\mathbf{q} \in \mathbb{H}$, $\|\mathbf{q}\| = 1$ represent orientation. $\otimes$ denotes quaternion multiplication and $(\cdot)^*$ the conjugate. A 3-vector $\mathbf{v}$ embedded as a pure quaternion is written $\bar{\mathbf{v}} = [0, \mathbf{v}^\top]^\top$. Sensor measurements in the body frame at step k are gyroscope $\boldsymbol{\omega}_k$, accelerometer $\mathbf{a}_k$, and magnetometer $\mathbf{m}_k$.

B. Optimization Stage 1: Euler Quaternion Integration

Standard VQF propagates orientation via the exact axis-angle formula:

$$\mathbf{q}_k^- = \mathbf{q}_{k-1} \otimes \begin{bmatrix} \cos \frac{\|\boldsymbol{\omega}_k\| \Delta t}{2} \\ \sin \frac{\|\boldsymbol{\omega}_k\| \Delta t}{2} \cdot \frac{\boldsymbol{\omega}_k}{\|\boldsymbol{\omega}_k\|} \end{bmatrix} \quad (1)$$

which requires a $\sqrt{\cdot}$, a $\cos$, and a $\sin$ per step. Fast VQF replaces this with the first-order Euler method applied to the quaternion kinematic equation $\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \bar{\boldsymbol{\omega}}$:

$$\mathbf{q}_k^- = \text{normalize}(\mathbf{q}_{k-1} + \frac{\Delta t}{2} \mathbf{q}_{k-1} \otimes \bar{\boldsymbol{\omega}}_k) \quad (2)$$

where $\bar{\boldsymbol{\omega}}_k = [0, \boldsymbol{\omega}_k^\top]^\top$. This requires only quaternion multiplication and addition, eliminating all transcendental calls. The approximation holds when $\|\boldsymbol{\omega}_k\| \Delta t \ll 1$, satisfied at typical IMU rates (≥ 100 Hz) for all but extreme angular velocities. Evaluation against both axis-angle and second-order Runge-Kutta integration confirms no measurable accuracy impact on the tested datasets. The gyroscope integration step itself runs approximately 15% faster; since this step accounts for roughly one third of the total pipeline, the net reduction in overall execution time is 5%. On microcontrollers without hardware floating-point units, where $\cos$, $\sin$, and $\sqrt{\cdot}$ are evaluated in software, the gyroscope integration step represents a larger fraction of total time and the overall gain from this optimization is expected to be proportionally greater.

C. Accelerometer Filtering in the Near-Earth Frame

This step is adopted unchanged from VQF [6]. The body-frame measurement is rotated into the gyroscope-propagated earth frame:

$$\tilde{\mathbf{a}}_k^g = \mathbf{q}_k^- \otimes \bar{\mathbf{a}}_k \otimes (\mathbf{q}_k^-)^* \quad (3)$$

A second-order Butterworth low-pass filter with cut-off frequency derived from time constant τ_{acc} is applied in this frame, yielding $\bar{\mathbf{a}}_k^g$, which suppresses linear-acceleration artefacts. The filtered vector is then rotated into the earth frame using the current correction quaternion $\mathbf{q}_{\text{acc}}$ and normalized:

$$\hat{\mathbf{a}}_k^{6D} = \frac{\mathbf{q}_{\text{acc}} \otimes \bar{\mathbf{a}}_k^g \otimes \mathbf{q}_{\text{acc}}^*}{\|\mathbf{q}_{\text{acc}} \otimes \bar{\mathbf{a}}_k^g \otimes \mathbf{q}_{\text{acc}}^*\|} \quad (4)$$

In VQF, $\mathbf{q}_{\text{acc}}$ is a tilt-only quaternion (roll and pitch, no heading). In Fast VQF, the same equation is applied with $\mathbf{q}_{\text{acc-mag}}$ in place of $\mathbf{q}_{\text{acc}}$, where $\mathbf{q}_{\text{acc-mag}}$ is the combined

tilt-and-heading correction quaternion defined in Optimization Stage 2 (Section III-D), so (4) rotates the filtered accelerometer into the full 9DOF earth frame rather than a tilt-only frame.

D. Optimization Stage 2: Combined Accelerometer-Magnetometer Correction

VQF correction structure: VQF maintains a tilt-correction quaternion $\mathbf{q}_{\text{acc}}$ and a scalar heading offset δ . The tilt correction computes the exact minimal rotation from $[0, 0, 1]^\top$ to $\hat{\mathbf{a}}_k^{6D}$:

$$\Delta \mathbf{q}_{\text{acc}}^{\text{VQF}} = \begin{bmatrix} \sqrt{\frac{\hat{a}_z + 1}{2}} \\ \frac{\hat{a}_y}{2} / \sqrt{\frac{\hat{a}_z + 1}{2}} \\ -\frac{\hat{a}_x}{2} / \sqrt{\frac{\hat{a}_z + 1}{2}} \\ 0 \end{bmatrix} \quad (5)$$

The heading correction computes the disagreement angle $\psi_k = \text{atan2}(m_x^e, m_y^e) - \delta$ and updates the scalar state via a first-order filter: $\delta_k = \delta_{k-1} + k_{\text{mag}} \psi_k$. This structure requires a $\sqrt{\cdot}$ in (5), an atan2 , and a separate δ state at every step.

Proposed combined correction: Fast VQF replaces both VQF corrections with a single linearized correction quaternion encoding the tilt update and a sign-based heading step simultaneously. Setting the scalar component to unity (valid when the per-step correction is small):

$$\mathbf{r}_k = \begin{bmatrix} 1 \\ \frac{1}{2} \hat{a}_y \\ -\frac{1}{2} \hat{a}_x \\ \sigma_k s_{\text{mag}} \end{bmatrix}, \quad \Delta \mathbf{q}_k = \frac{\mathbf{r}_k}{\|\mathbf{r}_k\|} \quad (6)$$

where $\hat{a}_x, \hat{a}_y$ are from $\hat{\mathbf{a}}_k^{6D}$, $s_{\text{mag}} = \tau_{\text{mag}} c$ is a fixed yaw step size (with constant c), and $\sigma_k = \text{sign}(m_x^{6D})$ is the sign of the magnetometer X-component in the 6D earth frame. The 6D earth frame has its Z axis aligned with gravity (determined by $\mathbf{q}_{\text{acc-mag}}$) and its X axis aligned with the current heading reference. Because the reference direction is placed on X, only the first component of the projected magnetometer vector carries information about the sign of the heading error—the Y and Z components are not needed and are never computed:

$$m_x^{6D} = [1, 0, 0] R(\mathbf{q}_{\text{acc-mag}} \otimes \mathbf{q}_k^-) \mathbf{m}_k \quad (7)$$

Computing only the first row of the rotation matrix costs 3 multiplications and 2 additions, avoiding both the full 3×3 rotation and the atan2 required by VQF. The sign σ_k determines the direction of the incremental yaw correction; the fourth component of $\Delta \mathbf{q}_k$ then encodes a small yaw rotation of magnitude s_{mag} in the error-reducing direction. The correction is applied as a single left-multiply:

$$\mathbf{q}_{\text{acc-mag},k} = \text{normalize}(\Delta \mathbf{q}_k \otimes \mathbf{q}_{\text{acc-mag},k-1}) \quad (8)$$

and the 9DOF orientation is:

$$\mathbf{q}_{9\text{DOF},k} = \mathbf{q}_{\text{acc-mag},k} \otimes \mathbf{q}_k^- \quad (9)$$

The savings come from: (i) the $\sqrt{\cdot}$ in (5) is replaced by the unit approximation, (ii) atan2 and the first-order heading filter are replaced by a sign evaluation, and (iii) the scalar δ state is

eliminated. This stage contributes 44 % of total execution time reduction. The combined correction concept derives from [3]; the magnetometer correction formulation is inherited directly from that work, where convergence properties of the sign-based step are analysed. As an alternative, a linear correction proportional to the heading error magnitude was evaluated on the same datasets; the sign-based step yielded consistently lower RMSE by approximately 3 %, confirming it as the better choice in practice. Near-zero oscillation of the sign step is not a practical concern: the fixed step magnitude s_{mag} is chosen to be well below the noise floor of typical MEMS magnetometers, so any residual switching is smaller than the inherent measurement noise.

IV. DATASET AND EVALUATION METHODOLOGY

A. Datasets

Evaluation uses two public benchmark datasets. The **BROAD** dataset [1] provides a large collection of takes from diverse motion scenarios and multiple sensor units, with an optical motion capture system as ground-truth reference. BROAD takes include both undisturbed takes (no significant magnetic or acceleration disturbance) and disturbed takes (recorded under magnetic disturbance or dynamic acceleration). In practice, all disturbed BROAD takes were also found to have optical reference discontinuities and are therefore classified as non-whitelist from a reference quality perspective. Readers interpreting results on these takes should be aware that the reference signal quality is compromised, and any observed accuracy differences could partly reflect reference artefacts. The **Sassari** dataset [2] contains three takes recorded at slow, medium, and fast motion speeds. Six IMU sensors were mounted simultaneously on a common platform; only sensor 0 is used in the present experiments. All three Sassari takes pass the whitelist criteria.

B. Optical Reference Discontinuities

Inspection of the BROAD reference signal reveals sporadic large discontinuities—sudden orientation jumps inconsistent with the physical motion—caused by marker occlusion or tracking loss in the optical system. These jumps introduce a persistent phase offset that incorrectly penalises all filter estimates equally.

Fig. 1 shows a representative example. Corrupted segments are identified by thresholding the frame-to-frame angular change in the reference signal. A 20s window following each detected jump is also excluded to prevent residual offset contamination. Segments passing both checks are reference-valid and are denoted the *whitelist*; in the present datasets these correspond to the undisturbed takes.

C. Magnetic Disturbance Segments

The BROAD dataset includes takes recorded in the presence of magnetic disturbance from ferromagnetic sources. In practice, these takes were found to also contain optical reference discontinuities, so they are included in the non-whitelist set and reported in analysis Section V-C.

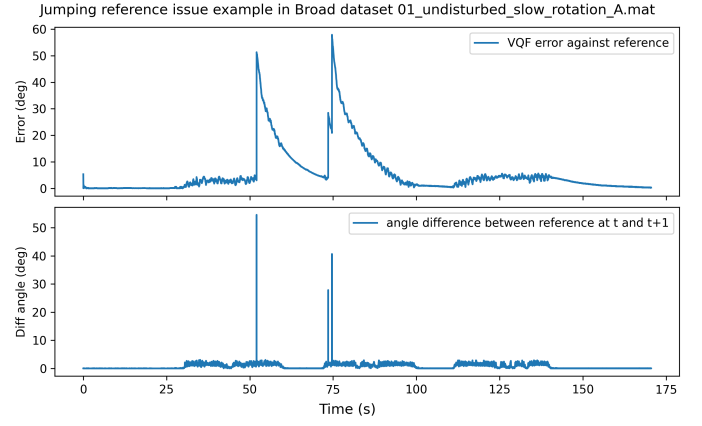


Fig. 1. Optical reference discontinuity in the BROAD dataset. A sudden jump in the reference (bottom) produces a long-term offset that affects all filter error metrics equally and must be excluded before fair comparison.

D. Evaluation Metrics

Orientation error is the RMSE of the geodesic angle between estimated and reference quaternions. Results are reported for the *undisturbed* takes and the *disturbed* takes. Computation time is the mean wall-clock time of the isolated filter update loop in a native optimized C++ build, measured on an AMD Ryzen 7 8700G desktop CPU. Timing measurements exhibit run-to-run variation of a few percent due to CPU scheduling and cache effects; each experiment was repeated many times and the mean is reported. The absolute timing values should therefore be treated as approximate, and comparisons between methods are more meaningful than the absolute figures. For each recording, the mean gyroscope reading during the initial static phase is subtracted before processing; all algorithms therefore run on gyroscope pre-calibrated sensor data. Since each take begins with a static phase, the reference quaternion is aligned to the filter estimate at the midpoint of that static phase; this eliminates the influence of different filter initialisation strategies and removes any fixed offset arising from a different choice of reference coordinate system.

E. Parameter Tuning

The accelerometer and magnetometer gain weights (τ_{acc} , τ_{mag} or s_{mag}) of Fast VQF were jointly optimised using the Nelder-Mead simplex method [7] to minimise mean orientation RMSE over the complete dataset, including both undisturbed and disturbed takes. The reported accuracy figures therefore represent training-set performance. Overfitting is not a significant concern here: only two scalar parameters are tuned, and the robustness of equivalent gain parameters to dataset variation has been established in the original VQF [6] and Justa AHRS [3] evaluations, which demonstrated that a single fixed parametrisation generalises well across diverse hardware and motion conditions.

TABLE I
COMPUTATION TIME AND ORIENTATION RMSE

Method	Time (ns)	Rel. [%]	RMSE _{undist} [°]	RMSE _{dist} [°]
Madgwick [4]	71.40	82.0	2.87	4.24
Justa AHRS [3]	64.91	74.6	2.84	3.96
VQF [6]	315.52	364.8	2.66	1.98
Basic VQF [6]	165.01	189.6	2.71	2.42
Fast VQF (ours)	87.02	100	2.62	2.92

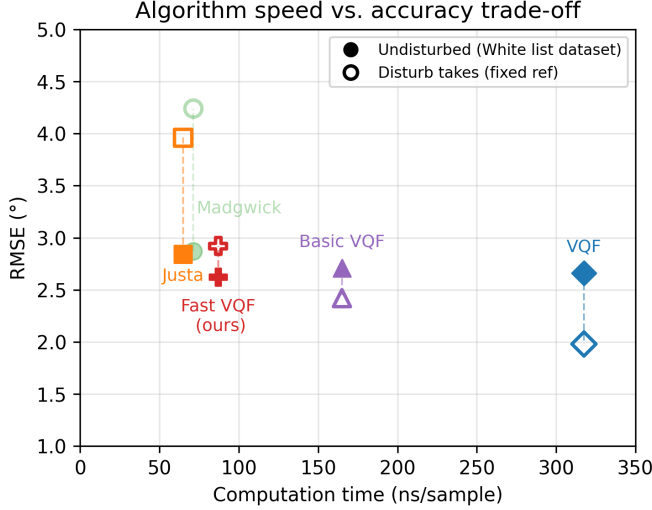


Fig. 2. Speed–accuracy trade-off for all evaluated methods on undisturbed data. Fast VQF occupies the best accuracy position among fast methods; full VQF provides additional benefit only under disturbance.

V. EXPERIMENTAL RESULTS

A. Speed and Accuracy

Table I reports computation time and orientation RMSE for all evaluated methods. The two Fast VQF optimization stages contribute independently: Euler integration reduces total execution time by 5 % relative to Basic VQF; the combined correction reduces it by a further 44 %. The net result is Fast VQF running at 87.02 ns per update— $\approx 1.9\times$ faster than Basic VQF (165.01 ns) and $\approx 3.8\times$ faster than full VQF (315.52 ns). Fig. 2 visualises the speed–accuracy trade-off for all methods.

On the undisturbed subset, Fast VQF achieves 2.62°—the lowest RMSE of all tested methods, 0.04° below full VQF (2.66°) and 0.09° below Basic VQF (2.71°). Justa AHRS [3] is the fastest method overall (64.91 ns) but yields 2.84° on undisturbed data, 0.22° above Fast VQF. Madgwick [4] is outperformed by both Justa AHRS and Fast VQF on every metric—neither fastest nor most accurate—confirming that it has been superseded by modern alternatives. The goniometric functions eliminated by Fast VQF are computed in software on microcontrollers without hardware floating-point units, so the speedup on such targets is expected to exceed the desktop figures.

The disturbed subset tells a different story. VQF’s gyroscope bias estimation and magnetic disturbance rejection provide a clear advantage under harsh conditions: 1.98° versus 2.92°

Filter Performance Across Undisturbed Takes

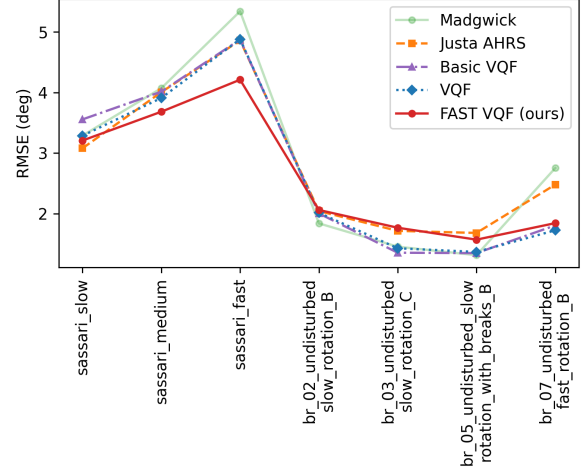


Fig. 3. Per-environment orientation RMSE on the undisturbed takes: Sassari slow/medium/fast (sensor 0) and BROAD undisturbed takes (all reference-valid).

for Fast VQF (gap of 0.94°, compared to a gap of -0.04° on undisturbed data). Justa AHRS and Madgwick degrade significantly on disturbed data (3.96° and 4.24°), confirming that algorithmic complexity is justified precisely in harsh environments, while fast methods are well-suited to undisturbed operation where computational cost and latency are the limiting factors.

B. Multi-Environment Accuracy

Fig. 3 shows per-environment RMSE on the undisturbed takes. Fast VQF maintains accuracy comparable to Basic VQF across all environments, confirming that the two optimizations do not introduce environment-specific failure modes.

C. Performance on Disturbed Data

Fig. 4 shows per-environment RMSE on the BROAD disturbed takes. Isolating the disturbed BROAD takes makes algorithm differences on hard conditions visible without cleaner takes masking the effect. Fast VQF’s sign-based heading correction provides behaviour comparable to VQF’s atan2 -based correction under moderate disturbance: the fixed step size s_{mag} bounds the yaw update per step, and the inherited VQF accelerometer low-pass filtering in the near-earth frame provides the same tilt robustness as the full algorithm.

VI. CONCLUSION

Fast VQF was presented as a speed-optimized 9DOF AHRS derived from the VQF algorithm through two targeted structural optimizations. Euler quaternion integration replaces the axis-angle propagation, eliminating $\cos/\sin$ calls and contributing 5 % of total time savings. A combined accelerometer–magnetometer correction replaces VQF’s two sequential updates, eliminating atan2 , an intermediate quaternion state, and a scalar heading variable, contributing a further 44 %.

Filter Performance Across Disturbed Takes (BROAD fixed)

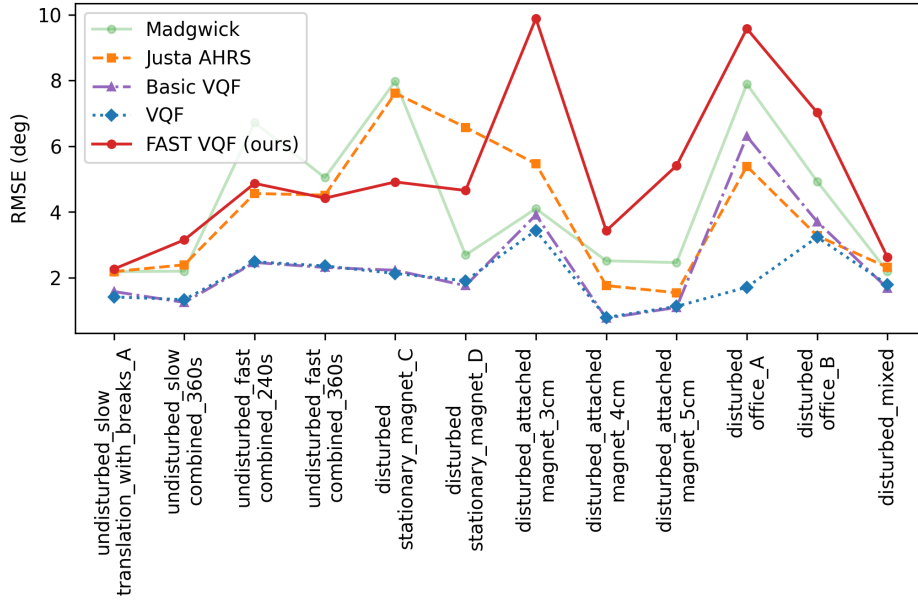


Fig. 4. Per-environment orientation RMSE on the BROAD disturbed takes (magnetic and acceleration disturbance; 20 s post-jump reference exclusion applied). Parameters were tuned on all takes including these.

Together, Fast VQF achieves approximately $1.9\times$ speedup over Basic VQF and $3.8\times$ over full VQF on desktop hardware, with an even larger gain expected on microcontrollers where trigonometric functions are computed in software. On undisturbed data, Fast VQF achieves the lowest RMSE of all tested methods (2.62°), outperforming Basic VQF (2.71°), full VQF (2.66°), Justa AHRS (2.84°), and Madgwick (2.87°). Under disturbed conditions, full VQF's additional mechanisms yield a clear advantage (1.98° vs. 2.92°). Evaluation against Justa AHRS and Madgwick confirms that modern fast methods are preferable to Madgwick on both speed and accuracy, and that Madgwick should no longer be the default choice for lightweight orientation estimation. Preprocessing analysis of the BROAD benchmark identified optical reference discontinuities that must be handled explicitly before valid filter comparison. Fast VQF is suited for resource-constrained embedded systems and latency-sensitive applications such as VR headset render-pose prediction, where algorithmic delay directly degrades prediction accuracy.

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REFERENCES

- [1] D. Laidig, M. Caruso, A. Cereatti, and T. Seel, "Broad—a benchmark for robust inertial orientation estimation," *Data*, vol. 6, no. 7, p. 72, 2021.
- [2] M. Caruso, A. M. Sabatini, M. Knaflitz, M. Gazzoni, U. Della Croce, and A. Cereatti, "Orientation estimation through magneto-inertial sensor fusion: A heuristic approach for suboptimal parameters tuning," *IEEE Sensors Journal*, vol. 21, no. 3, pp. 3408–3419, 2020.
- [3] J. Justa, V. Šmídl, and A. Hamáček, "Fast AHRS filter for accelerometer, magnetometer, and gyroscope combination with separated sensor corrections," *Sensors*, vol. 20, no. 14, p. 3824, 2020.
- [4] S. O. Madgwick *et al.*, "An efficient orientation filter for inertial and inertial/magnetic sensor arrays," 2010.
- [5] R. Mahony, T. Hamel, and J.-M. Pflimlin, "Nonlinear complementary filters on the special orthogonal group," *IEEE Transactions on automatic control*, vol. 53, no. 5, pp. 1203–1218, 2008.
- [6] D. Laidig and T. Seel, "VQF: Highly accurate IMU orientation estimation with bias estimation and magnetic disturbance rejection," *Information Fusion*, vol. 91, pp. 187–204, 2023.
- [7] J. A. Nelder and R. Mead, "A simplex method for function minimization," *The computer journal*, vol. 7, no. 4, pp. 308–313, 1965.